

CS280A — Project 2

Part 1: Fun with Filters Part 2: Fun with Frequencies

Part 1: Fun with Filters

We implement convolutions from scratch, finite differences for edges, Derivative of Gaussian (DoG), and a HSV orientation visualization.

1.1 Convolutions from Scratch

1. Zero padding helper and two convolution variants (`four_loops` & `two_loops`); compare to `scipy.signal.convolve2d`.
2. Test with a 9x9 box filter; report max absolute diff vs. SciPy.

Box blur (9x9) — four loops



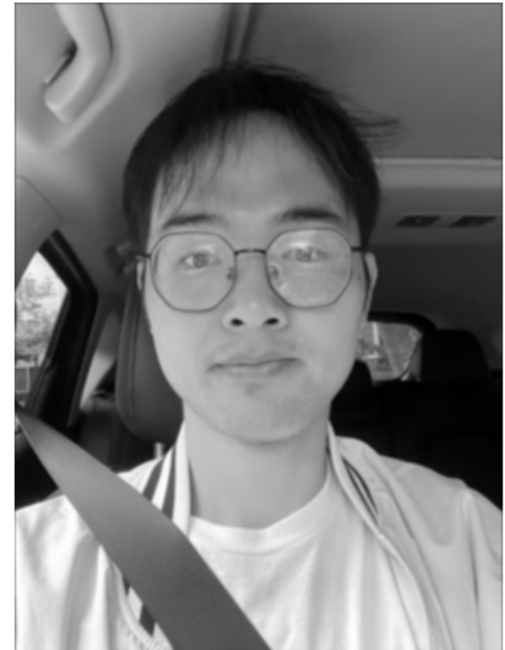
Box blur — four loops

Box blur (9x9) — two loops



Box blur — two loops

Box blur (9x9) — `scipy.signal.convolve2d`



Box blur — SciPy

Box blur (9x9) — four loops



Box blur — four loops

Box blur (9x9) — two loops



Box blur — two loops

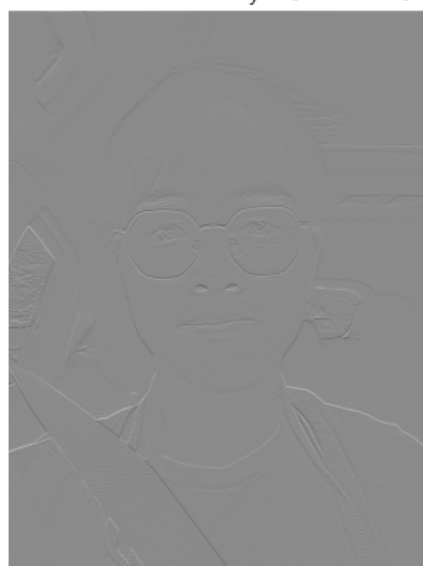
Box blur (9x9) — scipy.signal.convolve2d



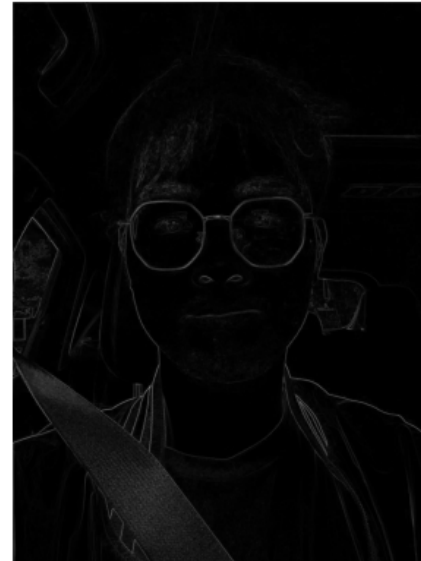
Box blur — SciPy

1.2 Finite Difference Operator

Use $\nabla_x = [1, 0, -1]$ and $\nabla_y = [0, 1, -1]^T$ to compute partial derivatives, gradient magnitude, and a percentile-based edge map.

Partial derivative $D_x = [1, 0, -1]$ Partial derivative ∇_x Partial derivative $D_y = [0, 1, -1]^T$ Partial derivative ∇_y

Gradient magnitude



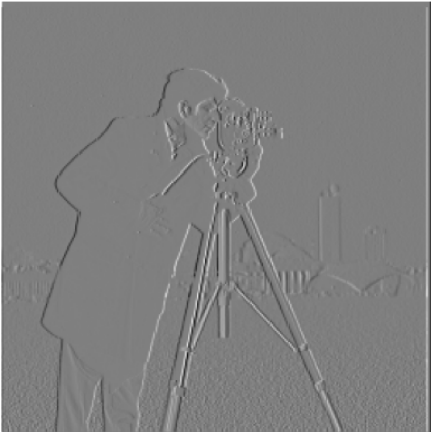
Gradient magnitude

Binary edge map (threshold = 85th percentile)



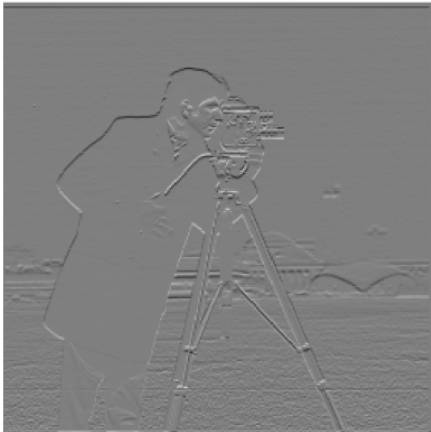
Binary edge map

Partial derivative I_x ($D_x * I$)



Partial derivative $\nabla(I_x)$

Partial derivative I_y ($D_y * I$)



Partial derivative $\nabla(I_y)$

Gradient magnitude



Gradient magnitude

Binary edge map (threshold = 85th percentile)



Binary edge map

1.3 Derivative of Gaussian (DoG)

Compare two pipelines: (A) blur → finite differences vs. (B) convolve with DoG kernels once. They produce similar results but (B) is a single pass.

Gaussian-blurred image



Gradient magnitude after Gaussian smoothing (blur → finite differences)



Grad magnitude after smoothing

Gradient magnitude via single DoG convolution



Grad magnitude via DoG

Gaussian-blurred image

Gaussian-blurred image



Gaussian-blurred image

Gradient magnitude after Gaussian smoothing (blur $\rightarrow$ finite differences)



Grad magnitude after smoothing

Gradient magnitude via single DoG convolution

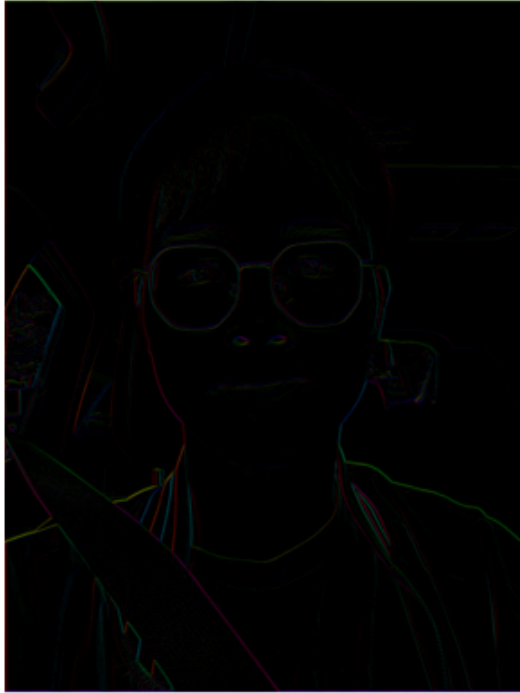


Grad magnitude via DoG

Bells & Whistles — HSV Orientation

Hue encodes orientation; value encodes gradient strength.

Gradient orientation (hue) + strength (value) [HSV → RGB]



Orientation (HSV→RGB)

Gradient orientation (hue) + strength (value) [HSV → RGB]



Orientation (HSV→RGB)

Part 2.1: Image “Sharpening”

Original Taj



Original

Gaussian Blur ($\sigma=1.8$)

Gaussian blur

Unsharp Masked (amount=1.2)



Unsharp masked

Unsharp via Single Convolution Kernel



Unsharp via single kernel

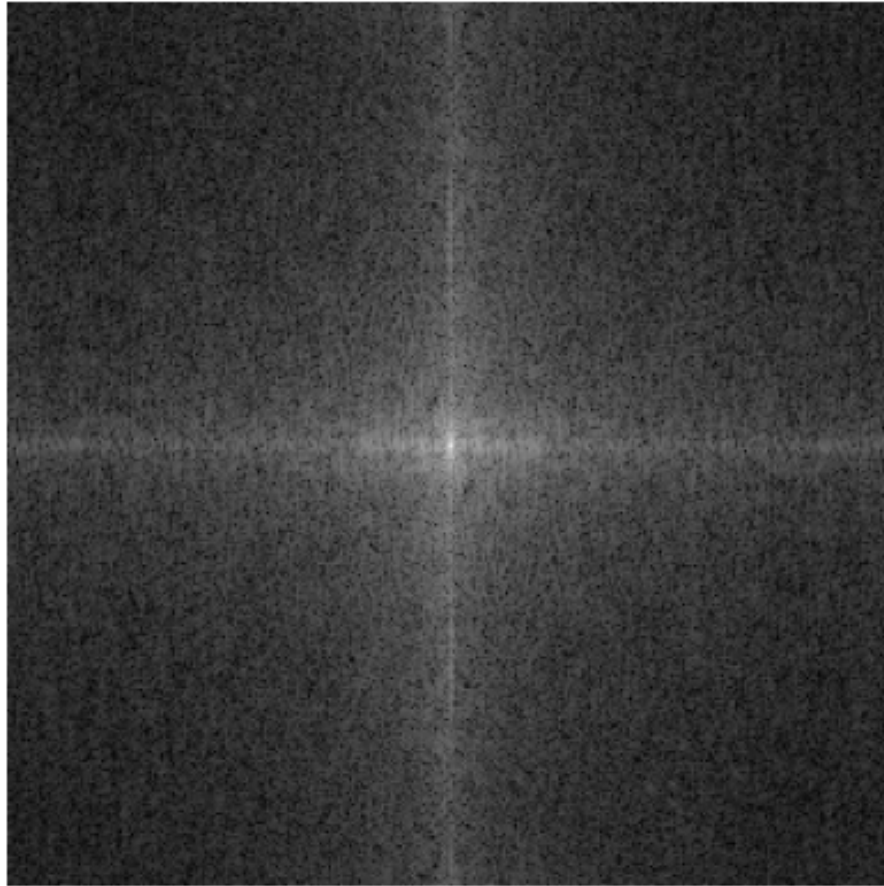
Sharpened Again



Sharpened again

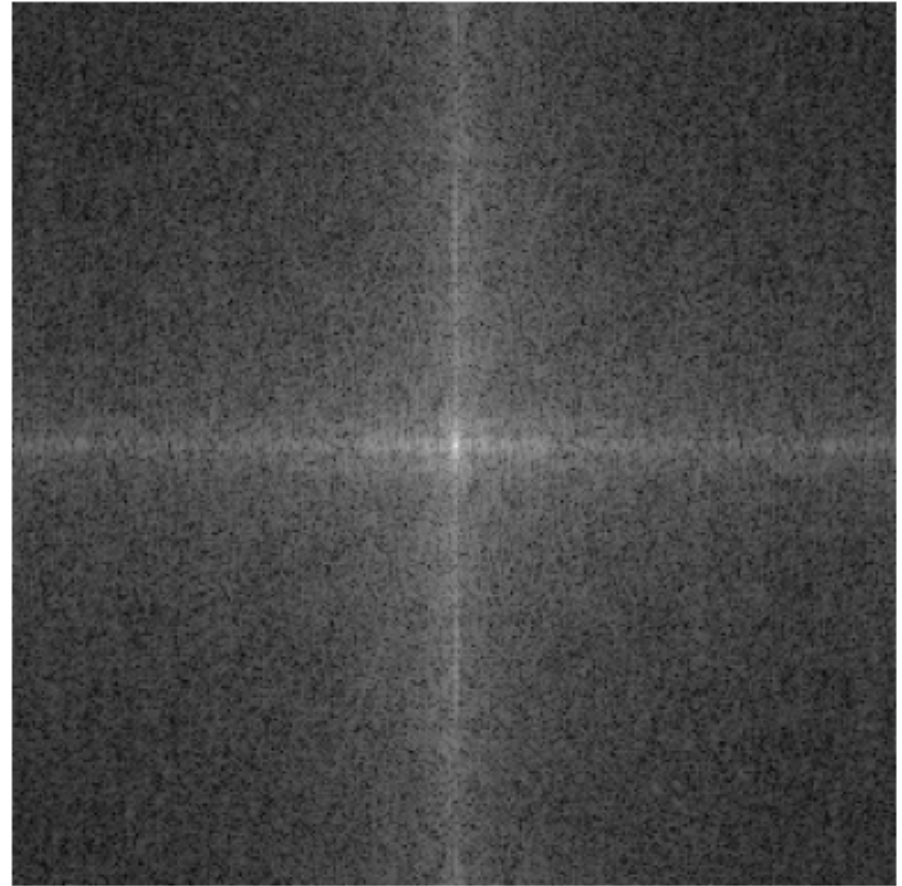
Frequency analysis (log |FFT|)

FFT (Taj)



FFT — original

FFT (Taj after Sharpening)



FFT — after sharpening

Part 2.2: Hybrid Images

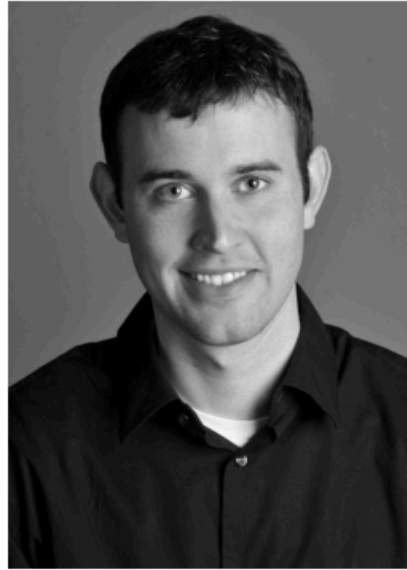
[Low-pass](#) [High-pass](#)

We find that adding color can help the low-frequency components.

Source (high-frequency image)



Target (low-frequency image)

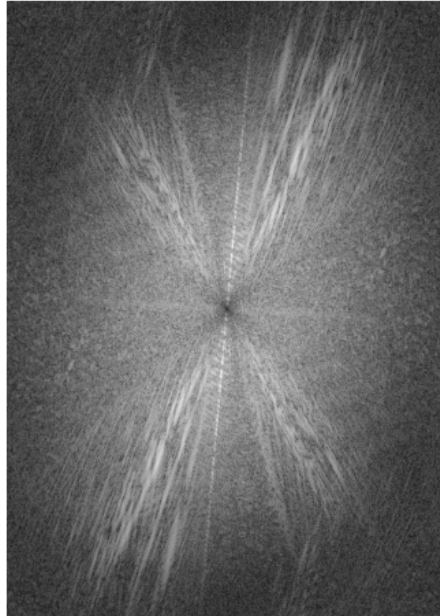
High-pass ($\sigma=2$)Low-pass ($\sigma=6$)

Hybrid

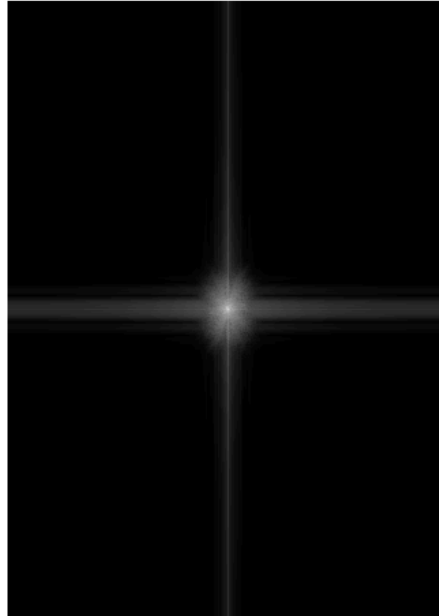


Low-frequency source (reference)

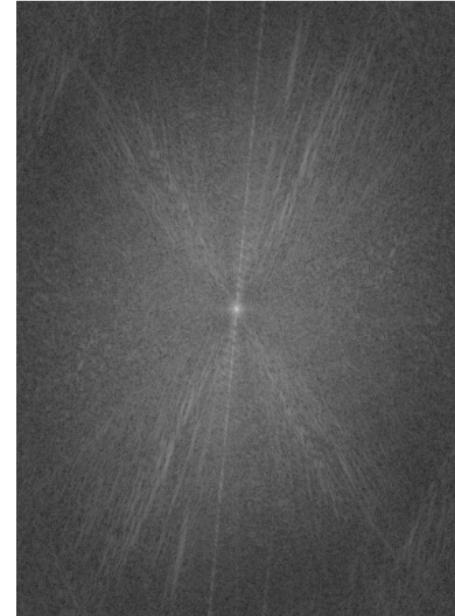
High-pass FFT



Low-pass FFT



Hybrid FFT



Low-frequency source (reference)

Source (high-frequency image)



Target (low-frequency image)

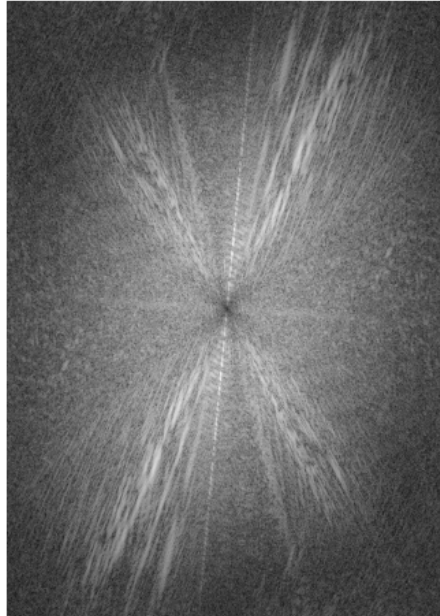
High-pass ($\sigma=2$)Low-pass ($\sigma=6$)

Hybrid

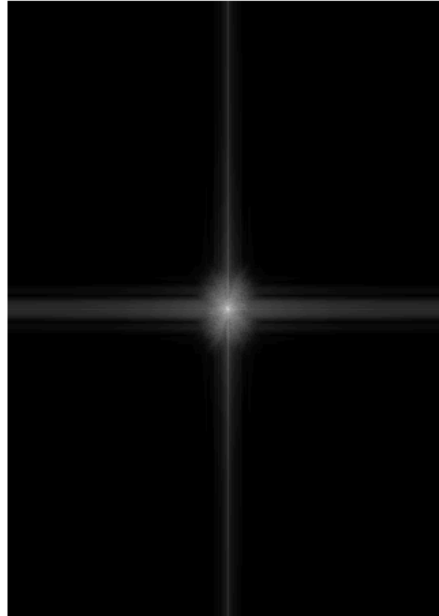


Low-frequency source with colors (reference)

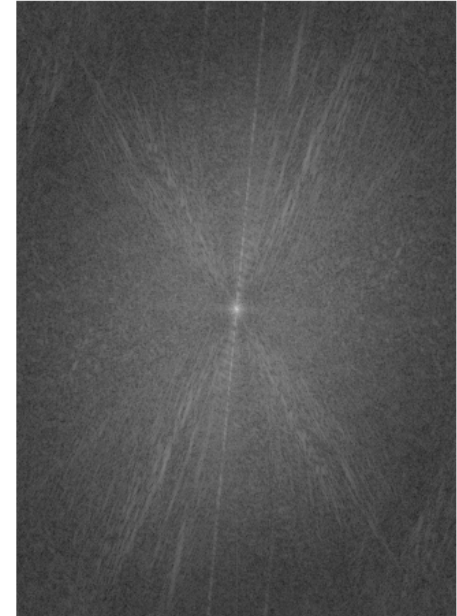
High-pass FFT



Low-pass FFT



Hybrid FFT



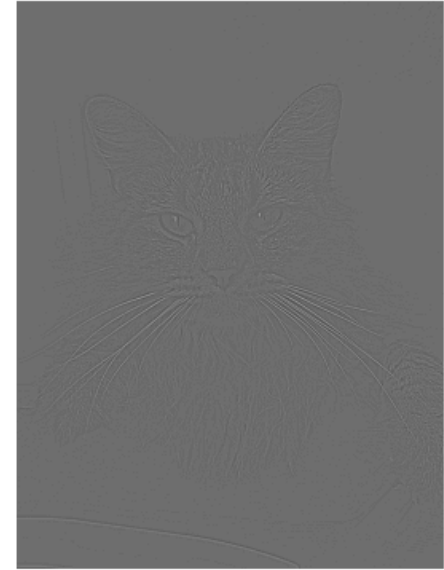
Low-frequency source with colors (reference)

[More Hybrid Examples](#)

Source (high-frequency image)



Target (low-frequency image)

High-pass ($\sigma=2$)Low-pass ($\sigma=6$)

Hybrid



Hybrid example #2

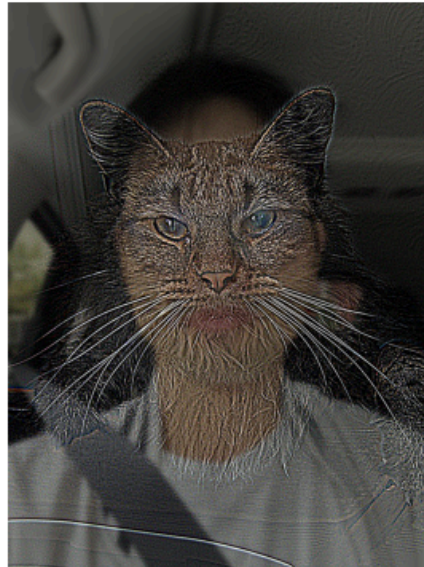
Source (high-frequency image)



Target (low-frequency image)

High-pass ($\sigma=2$)Low-pass ($\sigma=6$)

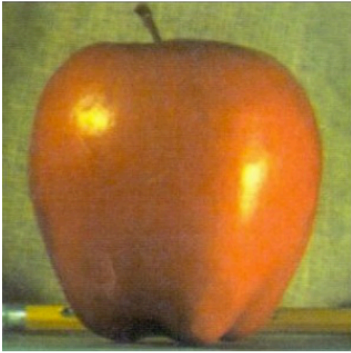
Hybrid



Hybrid example #3

Part 2.3: Gaussian and Laplacian Stacks

Gaussian A 0



Gaussian A 1



Gaussian A 2



Gaussian A 3



Gaussian A 4

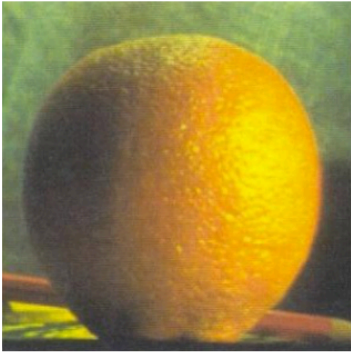


Gaussian A 5



Gaussian Stack for apples

Gaussian B 0



Gaussian B 1



Gaussian B 2



Gaussian B 3



Gaussian B 4

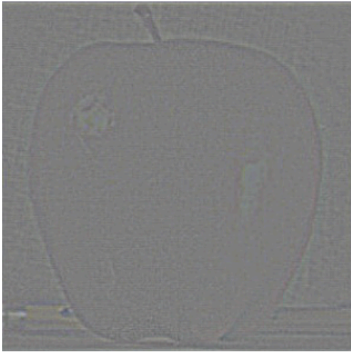


Gaussian B 5



Gaussian Stack for oranges

Laplacian A (+0.5) 0



Laplacian A (+0.5) 1



Laplacian A (+0.5) 2



Laplacian A (+0.5) 3



Laplacian A (+0.5) 4

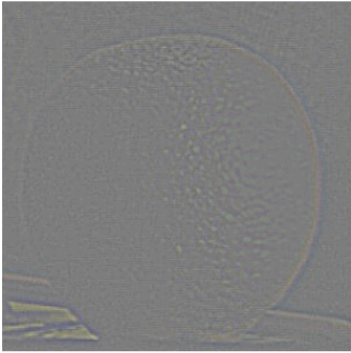


Laplacian A (+0.5) 5



Laplacian Stack for apples

Laplacian B (+0.5) 0



Laplacian B (+0.5) 1



Laplacian B (+0.5) 2



Laplacian B (+0.5) 3



Laplacian B (+0.5) 4



Laplacian B (+0.5) 5



Laplacian Stack for oranges

Image A

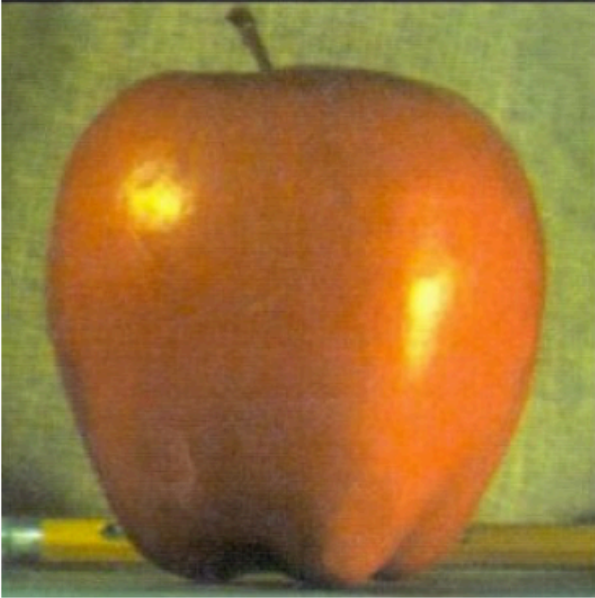
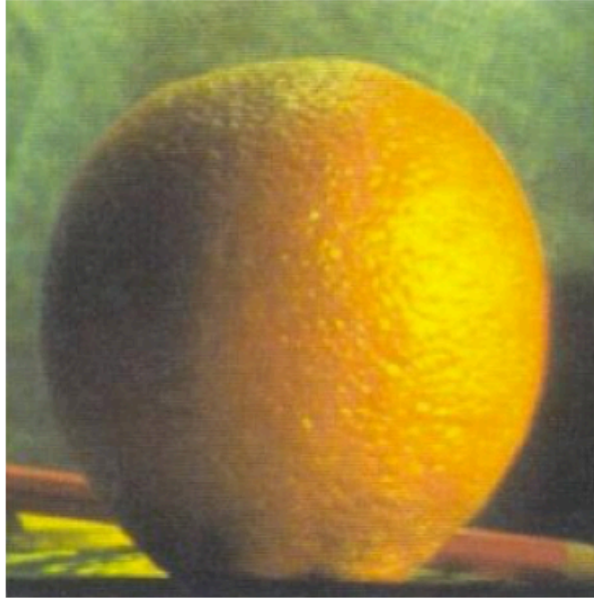
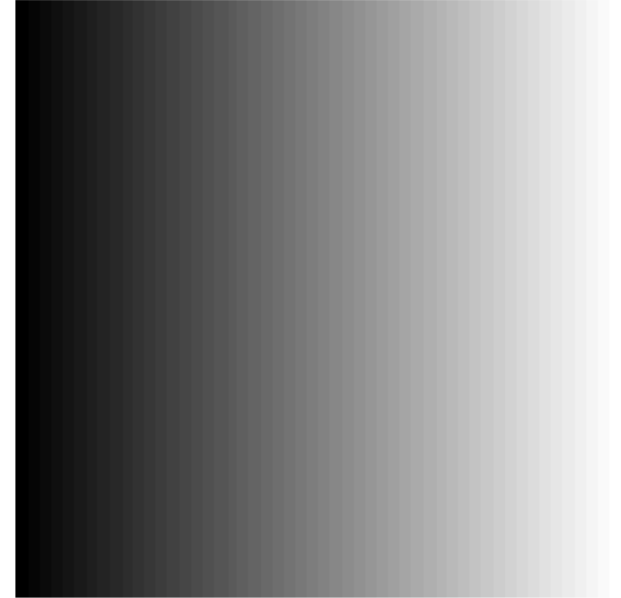


Image B



Mask (white→A, black→B)

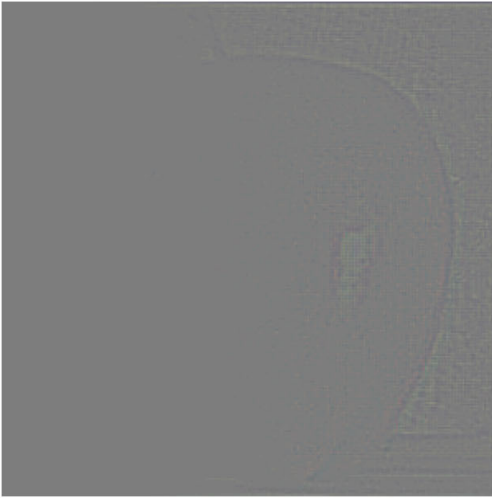


Blending procedures

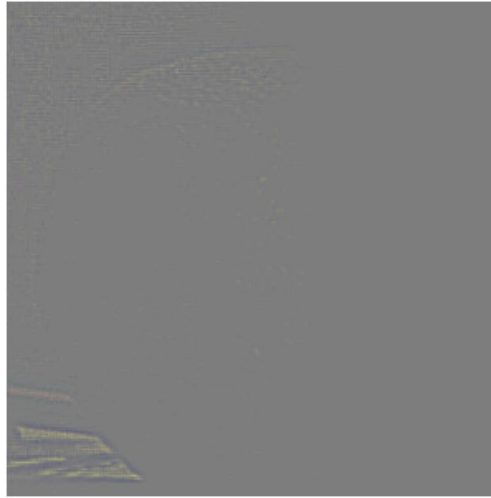
Multiresolution Blend (Orapple)



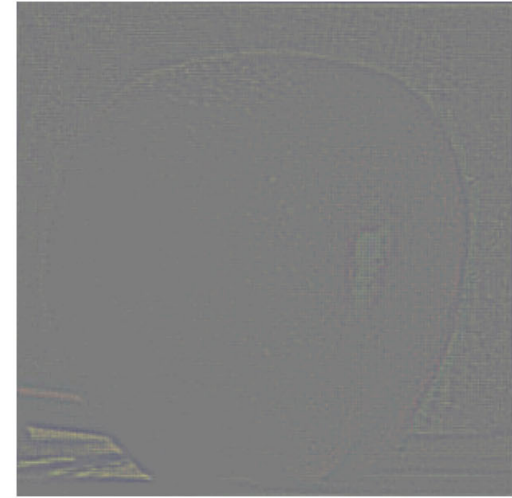
(a)



(b)



(c)



(d)



(e)



(f)



Image A

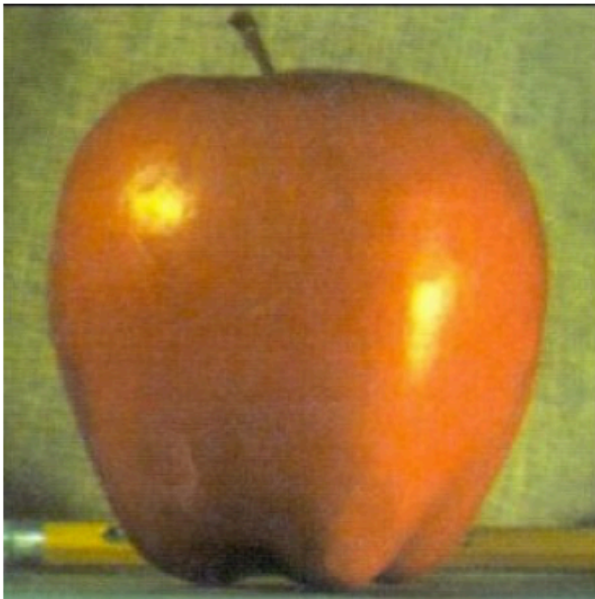
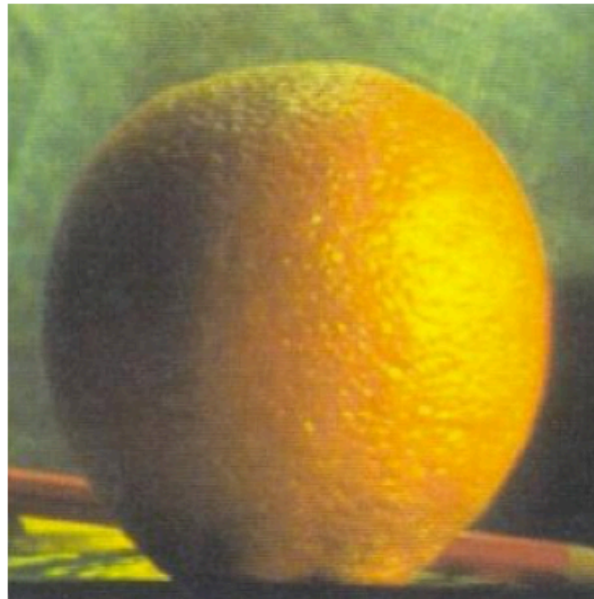
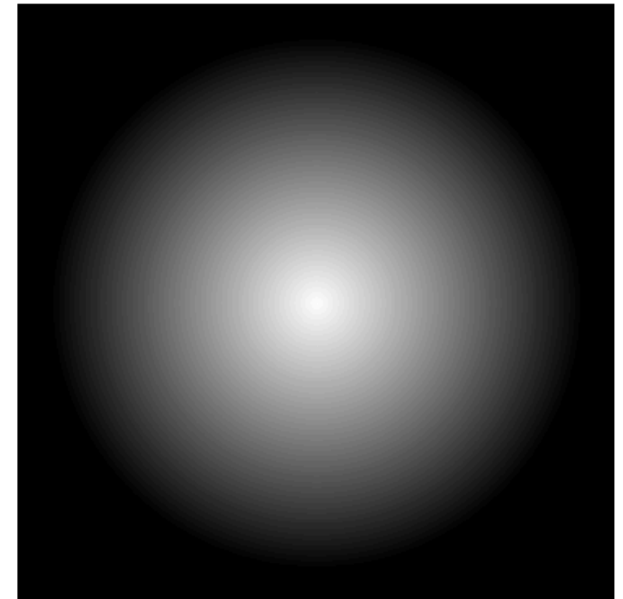


Image B



Mask (white→A, black→B)



Blending procedure with circle mask



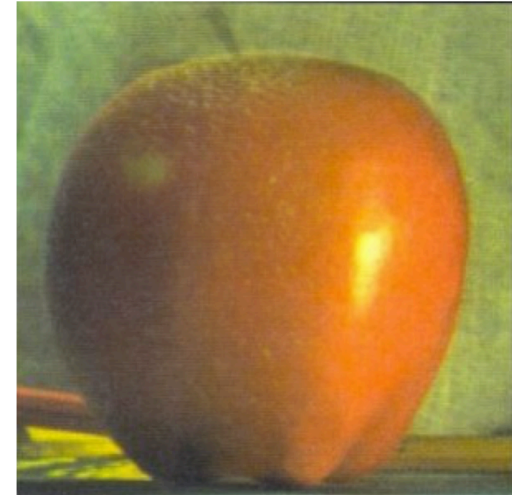
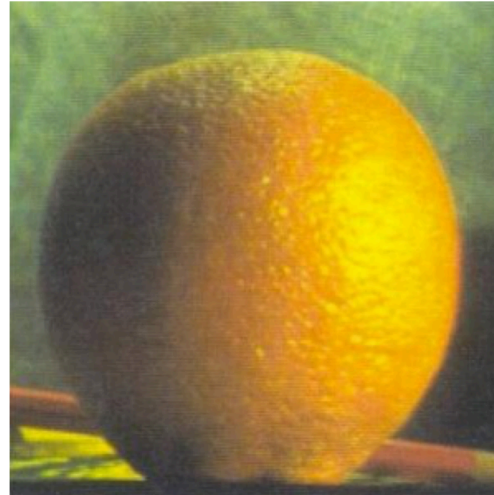
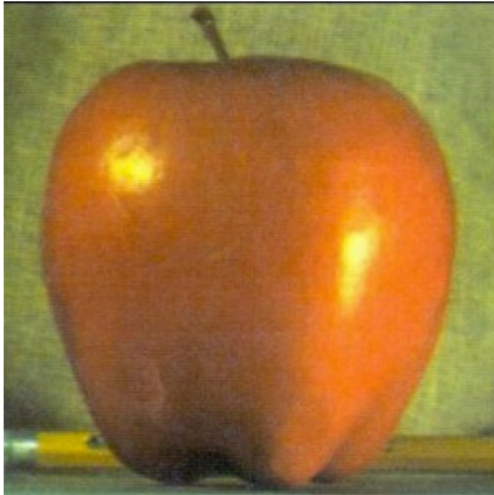
(j)



(k)



(l)



Multiresolution Blend (Orapple)



